$$\sin(\alpha+\beta)-\sin(\alpha-\beta)=2\cos\alpha\sin\beta$$

Let $\alpha = rx, \quad \beta = \dfrac{x}{2}$

$$\sin\left(r+\frac{1}{2}\right)x-\sin\left(r-\frac{1}{2}\right)x=2\sin\frac{x}{2}\cos rx$$

$$\begin{aligned}2\sin\frac{x}{2}\left(\cos x+\cos 2x+\cdots+\cos nx\right)&=\sum_{r=1}^{n}\left(\sin\left(r+\frac{1}{2}\right)x-\sin\left(r-\frac{1}{2}\right)x\right)\\&=\sin\left(n+\frac{1}{2}\right)x-\sin\frac{x}{2}\end{aligned}$$

$$\therefore \cos x+\cos 2x+\cdots+\cos nx=\frac{\sin\left(n+\frac{1}{2}\right)x-\sin\frac{x}{2}}{2\sin\frac{x}{2}}$$

(i)

$$S_2(x)=\sin x+\frac{1}{2}\sin 2x$$

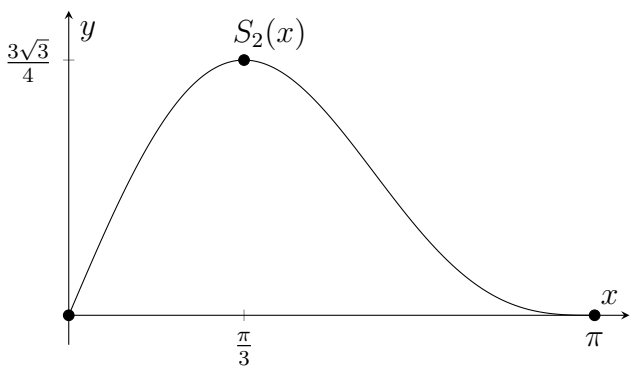
$$S_2'(x)=\cos x+\cos 2x$$

$$S_2'(x)=0 \implies \cos x+\cos 2x=0$$

$$\cos x+2\cos^2 x-1=0$$

$$(2\cos x-1)(\cos x+1)=0 \implies x=\frac{\pi}{3},\ \pi$$

x	$\frac{\pi}{3}$	π
y	$\frac{3\sqrt{3}}{4}$	0



(ii)

$$S_n'(x) = \sum_{r=1}^{n} \cos rx$$

$$\frac{\sin\left(n+\frac{1}{2}\right)x_0 - \sin\frac{x_0}{2}}{2\sin\frac{x_0}{2}} = 0$$

$$\sin\left(n+\frac{1}{2}\right)x_0 = \sin\frac{x_0}{2}$$

$$\sin nx_0 \cos\frac{x_0}{2} + \cos nx_0 \sin\frac{x_0}{2} = \sin\frac{x_0}{2}$$

$$\sin nx_0 \cos\frac{x_0}{2} = (1-\cos nx_0)\sin\frac{x_0}{2}$$

$$\sin nx_0 = (1-\cos nx_0)\tan\frac{x_0}{2}$$

$$S_n(x_0) - S_{n-1}(x_0) = \frac{1}{n}\sin nx_0 = \frac{1}{n}(1-\cos nx_0)\tan\frac{x_0}{2} \geq 0$$

$$\therefore S_n(x_0) \geq S_{n-1}(x_0)$$

If $S_n(x) \leq 0$ for some $0 < x < \pi$, let x_0 be a minimum point of $S_n(x)$ in $0 < x < \pi$.

$$S_n'(x_0) = 0 \implies S_n(x_0) \geq S_{n-1}(x_0) > 0$$

Contradiction.

$$\therefore S_n(x) > 0 \qquad (0 < x < \pi)$$

(iii)

$$S_1(x) = \sin x > 0 \qquad (0 < x < \pi)$$

$$S_{n-1}(x) > 0 \implies S_n(x) > 0 \qquad (0 < x < \pi)$$

$$\because S_n(0) = S_n(\pi) = 0$$

$$\therefore S_n(x) \geq 0 \qquad (0 \leq x \leq \pi)$$

This solution was submitted by **Jon**
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